

Spotify

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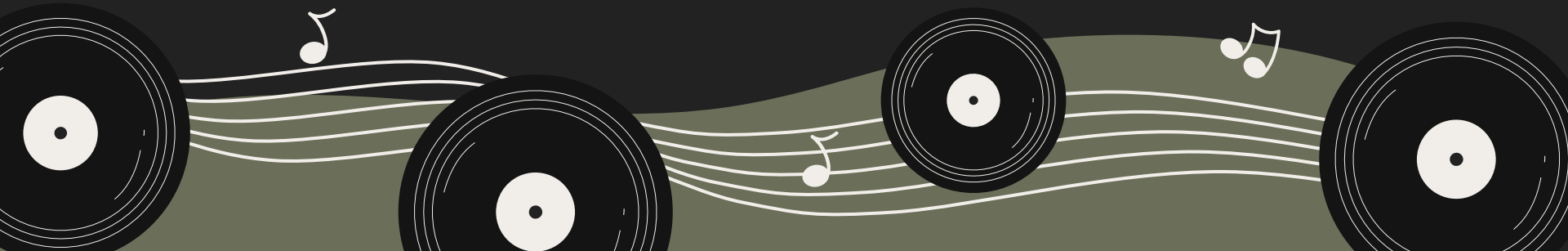




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01

Introduction of Problem

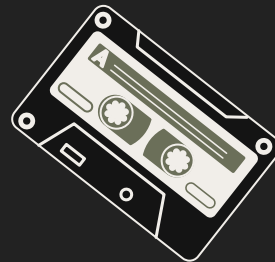


Introduction of Problem



Topic One:

Impact of Popular
Songs



Topic Two:

Changes in Musical
Attributes





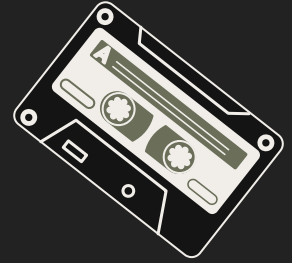
Impact of Popular Songs

How does the presence of one or more highly popular songs in an album affect the popularity and listener engagement of other tracks within the same album?

- Number of highly popular songs
- Presence of popular vs highly popular song
- Impact of popular collaboration v popular solo?
- Consistent for genres?



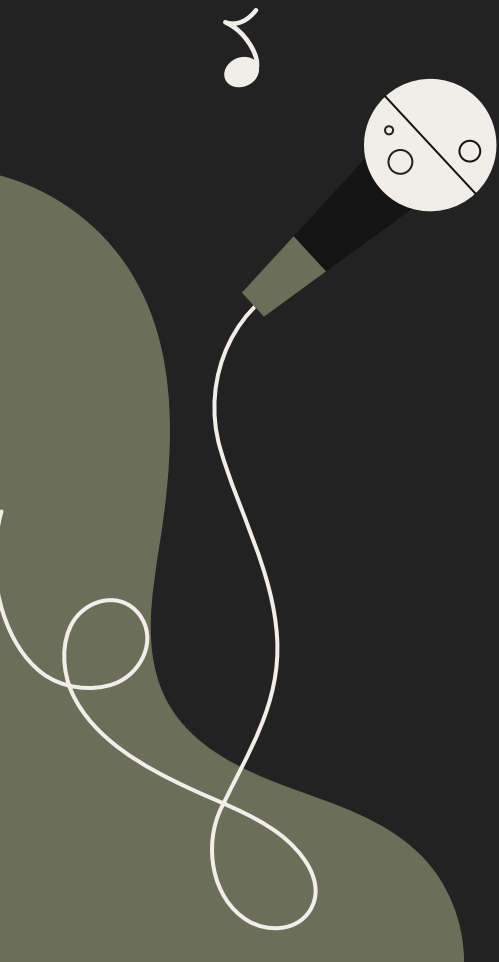
Changes in Musical Attributes



How do the musical attributes of tracks change when artists collaborate compared to when they work solo?

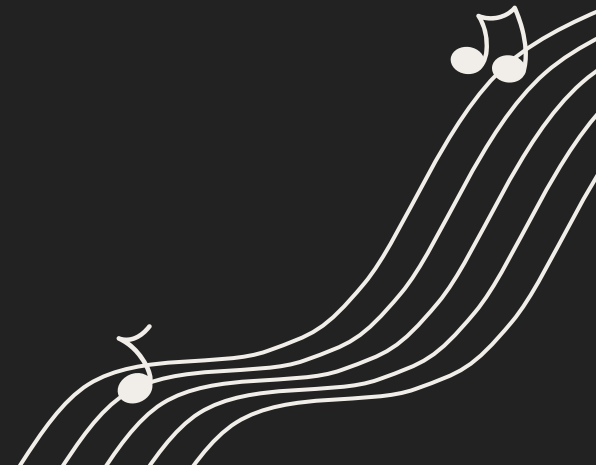
- Number of collaborators
- Different artist genres collaborate
- Attribute Relationship to popularity
- Attribute Relationship to collaborations





02

Data Analysis Step





Data Science Process



01

Capture



02

Maintain



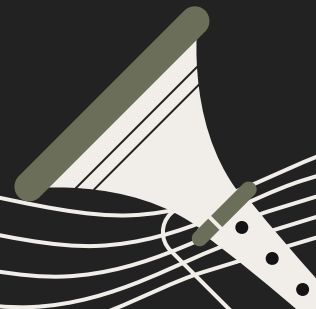
03

Analysis



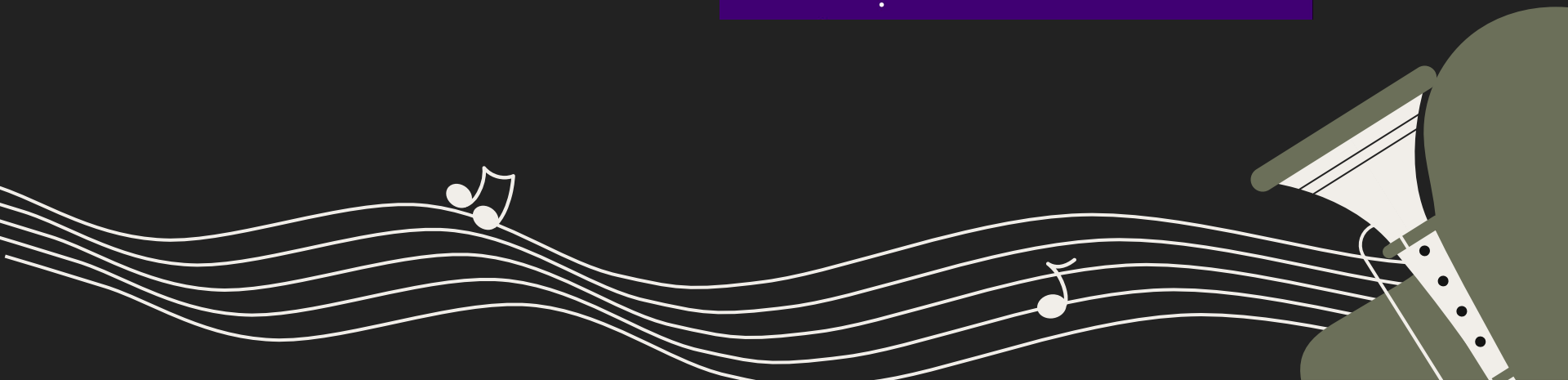
04

Communication



01- Capture

- Obtained data **directly** from Spotify API
 - Available with documentation



02 - Maintain



```
In [24]: #Creating popular_track variable
#HP > 80, N < 50
#50<= P < 80

def popular_tracks(popularity):
    if popularity >= 80:
        return 'HP'
    elif popularity >= 50:
        return 'P'
    else:
        return 'N'

df['popular_track'] = df['popularity'].apply(popular_tracks)

In [25]: #creates variables num_popular and num_high_popular
# counts the number of (highly) popular songs in an album

num_popular = df.groupby('album_name')['popular_track'].apply(lambda x: (x==
num_high_popular = df.groupby('album_name')['popular_track'].apply(lambda x:

df = df.merge(num_popular, on='album_name', how='left').merge(num_high_pc

In [26]: df['num_artist'] = df['artists'].apply(lambda x: len(x.split(';')))

In [27]: df['collab'] = df['num_artist'].apply(lambda x: 'C' if x > 1 else 'S')

In [28]: album_popularity = df.groupby('album_name')['popularity'].mean().reset_index
album_popularity = album_popularity.rename(columns={'popularity': 'album_avg
df = df.merge(album_popularity, on='album_name', how='left')
df.head()
```

```
Out[28]:
```

	track_id	artists	album_name	track_name	popularity
0	5SuOikwiRyPMVolQDJUGSV	Gen Hoshino	Comedy	Comedy	73
1	4qPNDBW1i3p13qLct0Ki3A	Ben Woodward	Ghost (Acoustic)	Ghost - Acoustic	55
2	1iJBS7s7jYXzMBEGcbK5b	Ingrid Michaelson;Zayn	To Begin Again	To Begin Again	57
3	6lfxq3CG4xtTIEg7opyCyx	Kina Grannis	Crazy Rich Asians (Original Motion Picture Sou...	Can't Help Falling In Love	71
4	5vjLSffimilP26QG5WcN2K	Chord Overstreet	Hold On	Hold On	82

5 rows x 27 columns

- Loading into Python
- Data Cleaning
 - Removing Duplicates
 - String Cleaning Methods
- Creating Variables
 - Indicators
 - Popularity
 - Collaborations
 - Albums



Impact of Popular Songs *Analysis*

Popularity (0-100):

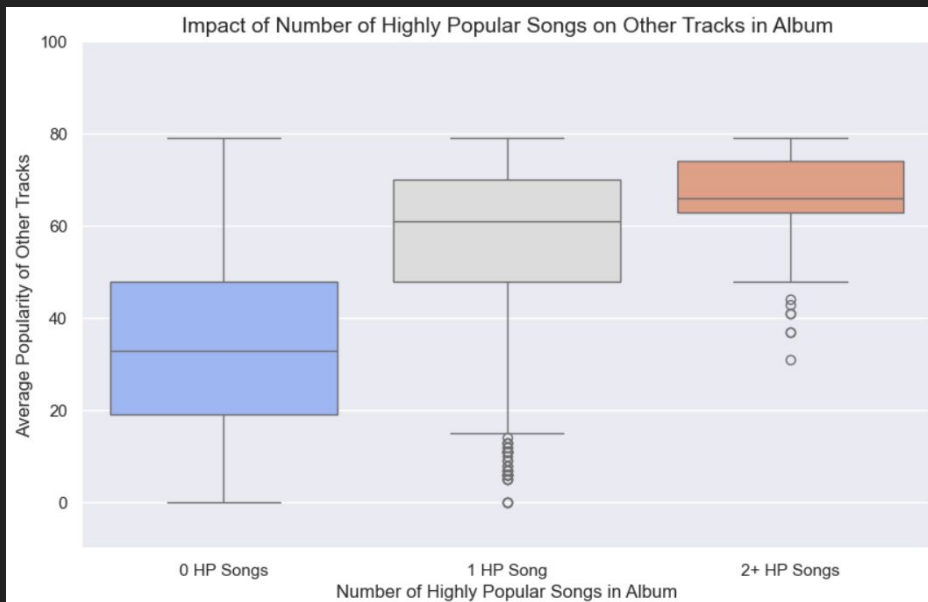
Not Popular (0-49), Popular (50-79), Highly Popular (80+)



03.1 - Analysis

Topic One: Impact of Popular Songs

Number of HP songs



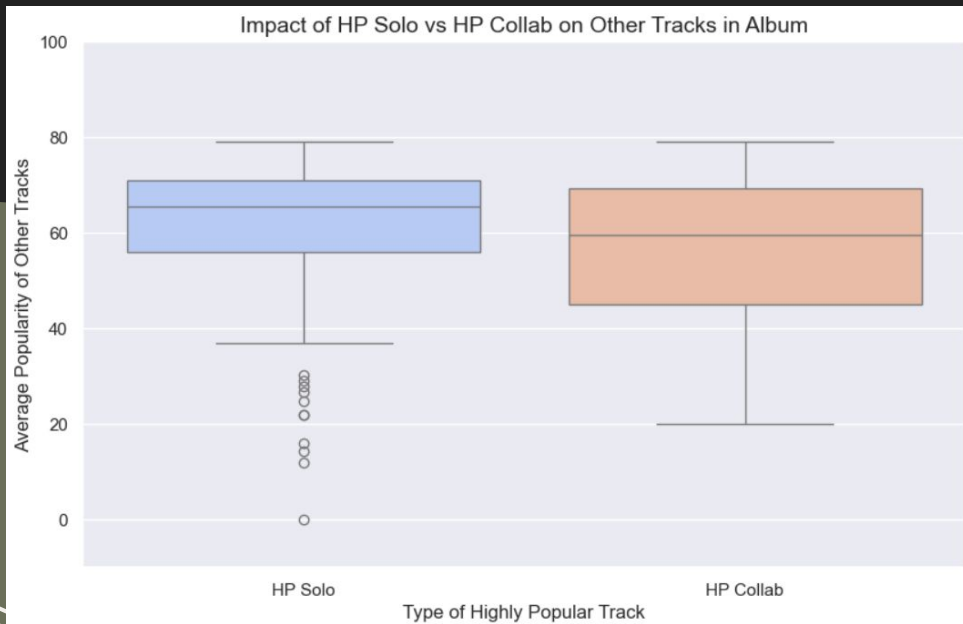
- Trend witnessed:
 - Albums with more HP songs **have higher** listener engagement + album popularity
 - Some overlap of box plots
- P-value = $4.4541e-254$



03.1 - Analysis

Topic One: Impact of Popular Songs

Impact of P collab v P solo



- Trend witnessed:
 - Solos albums **have slightly higher** engagement + album popularity
 - Some overlap of box plots
- P-value = 1.2505e-02



03.1 - Analysis

Topic One: Impact of Popular Songs

Impact of P collab v P solo



Album
The Magic Hour Collection
Jhené Aiko • 2024 • 25 songs, 1 hr 35 min

Play controls: Play, Repeat, Add, Download, More

#	Title	Plays	⌚
1	guidance Jhené Aiko	6,770,485	3:35
2	3:16AM Jhené Aiko	41,069,682	3:20
3	Stay Ready (What A Life) ... Jhené Aiko, Kendrick La...	284,559,611	6:22
4	Oblivion (Creation) [feat. ...] Jhené Aiko, Dr. Chill	10,741,887	5:50
5	While We're Young Jhené Aiko	313,103,810	3:56



Album
Sunburn
Dominic Fike • 2023 • 15 songs, 39 min 13 sec

Play controls: Play, Repeat, Checkmark, Download, More

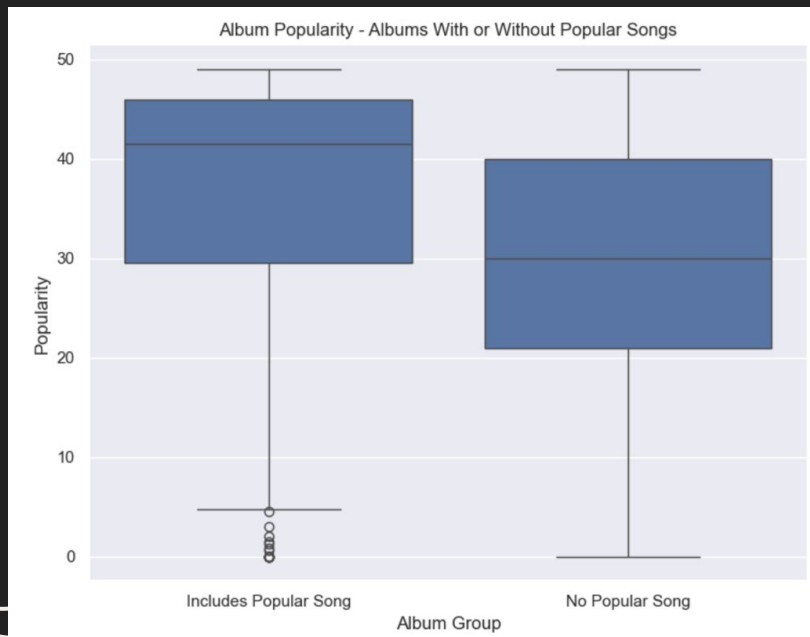
#	Title	Plays	⌚
1	How Much Is Weed? Dominic Fike	52,062,266	2:05
2	Ant Pile Dominic Fike	29,707,028	2:06
3	Think Fast (feat. Weezer) Dominic Fike, Weezer	34,068,194	3:42



03.1 - Analysis

Topic One: Impact of Popular Songs

Presence of P v HP Songs



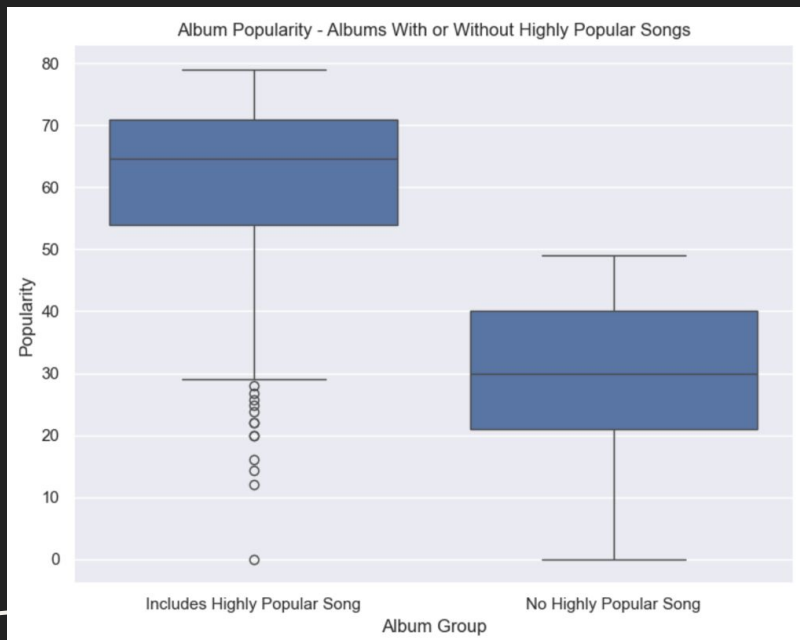
- Trend witnessed:
 - Album **without** popular songs lower engagement + album popularity
 - Some overlap of box plots
- P-value = $3.59e-230$



03.1 - Analysis

Topic One: Impact of Popular Songs

Presence of P v HP Songs



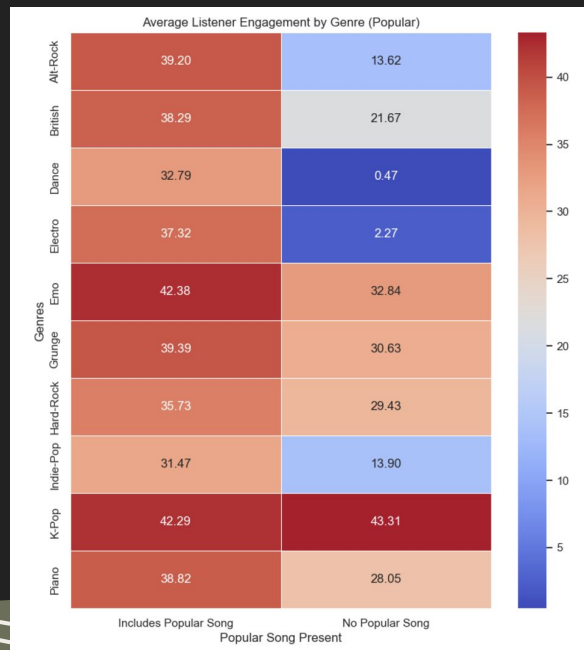
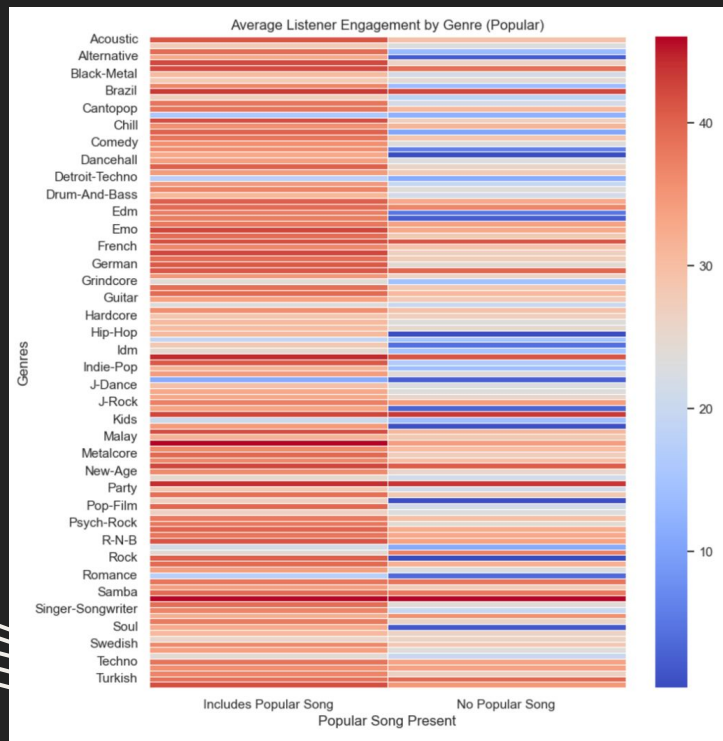
- Trend witnessed:
 - Album **without** highly popular songs, lower engagement + album popularity
 - No overlap of box plots
- P-value = $1.71e-83$



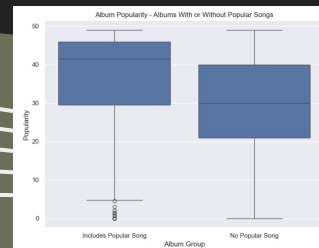
03.1 - Analysis

Topic One: Impact of Popular Songs

Genre Consistency? (P/HP)



- Genres:
 - Album **with** popular songs **HIGHER** engagement + album popularity (red)
 - Some overlap of red



For all genres:

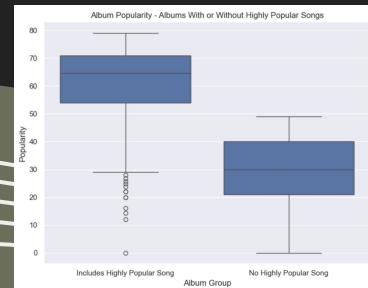
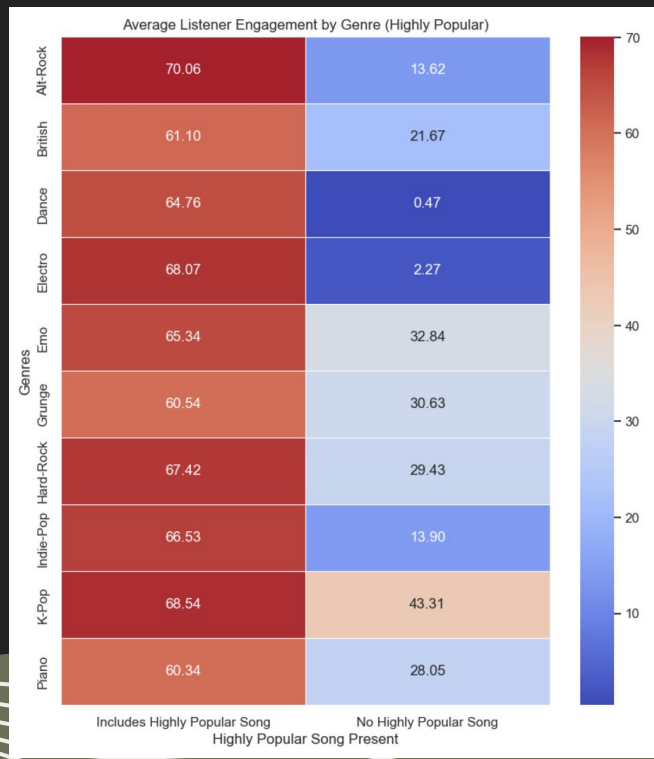
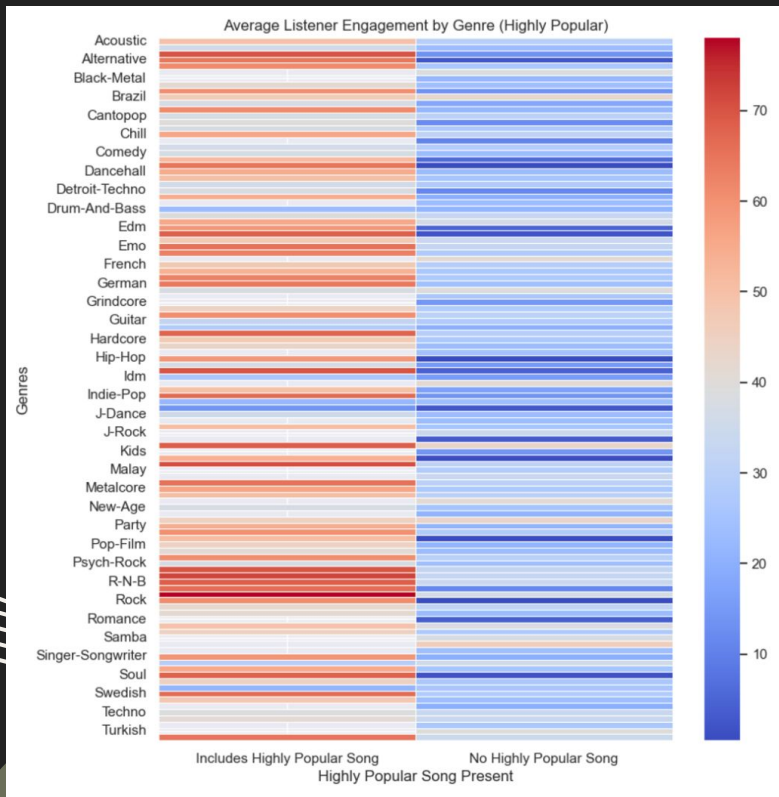
Small snippet: closer view



03.1 - Analysis

Topic One: Impact of Popular Songs

Genre Consistency? (HP)





03.1 - Analysis

Topic One: Impact of Popular Songs

Genre Consistency? (HP)

	Genre	t-statistic	p-value	Average Engagement (With HP)	Average Engagement (Without HP)	Significant
0	Acoustic	3.554755	2.143205e-04	45.741142	30.939823	True
1	Afrobeat	3.263166	5.924518e-04	38.677778	22.304982	True
2	Alt-Rock	8.472137	6.291256e-16	69.298232	26.061015	True
3	Alternative	16.697083	1.304702e-35	60.185096	2.650227	True
4	Ambient	3.997599	4.429508e-05	58.842593	33.305767	True
...	...	...	...	...	...	...
60	Study	1.291472	9.850276e-02	37.666667	28.364635	False
61	Swedish	3.594620	1.823041e-04	66.100000	31.850863	True
62	Synth-Pop	4.112742	2.424242e-05	50.828947	24.903388	True
63	Techno	0.892666	1.864671e-01	39.131579	33.223270	False
64	Trance	1.095209	1.370901e-01	40.394218	35.742175	False

	track_genre	group	count
0	Brazil	Includes Highly Popular Song	3
1	Brazil	No Highly Popular Song	567
2	Deep-House	Includes Highly Popular Song	6
3	Deep-House	No Highly Popular Song	300
4	Drum-And-Bass	Includes Highly Popular Song	2
5	Drum-And-Bass	No Highly Popular Song	558
6	Dub	Includes Highly Popular Song	2
7	Dub	No Highly Popular Song	429
8	Guitar	Includes Highly Popular Song	3
9	Guitar	No Highly Popular Song	493
10	Idm	Includes Highly Popular Song	2
11	Idm	No Highly Popular Song	438
12	Ska	Includes Highly Popular Song	2
13	Ska	No Highly Popular Song	420
14	Study	Includes Highly Popular Song	3
15	Study	No Highly Popular Song	640
16	Techno	Includes Highly Popular Song	3



03.1 - Analysis

Topic One: Impact of Popular Songs

Popular Singles in Albums (Duplicates)



Album
SOS
SZA • 2022 • 23 songs, 1 hr 8 min

Title

1	SOS SZA
2	Kill Bill SZA

Kill Bill
Single • 2023 • 4 songs

Title

1	Kill Bill SZA	2:33
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Album
You'll Be Alright, Kid (Chapter 1)
Alex Warren • 2024 • 10 songs, 30 min

Title

1	Burning Down Alex Warren
2	Catch My Breath Alex Warren

Burning Down
Single • 2024 • 1 song

Title

1	Burning Down Alex Warren	2:59
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03.1 - Analysis

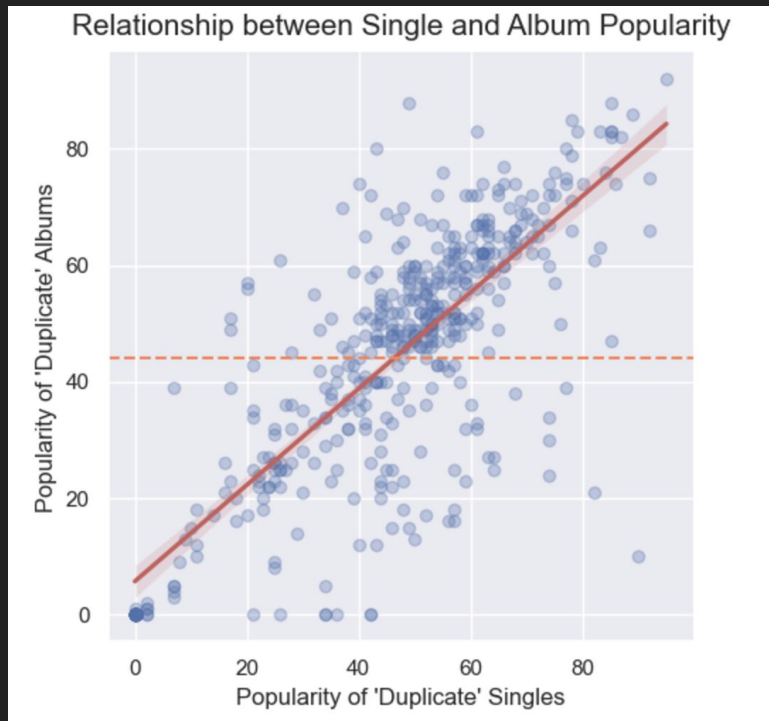
Topic One: Impact of Popular Songs

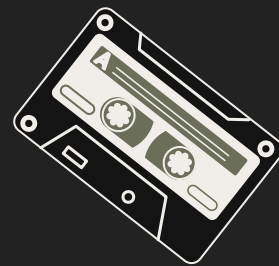
Popular Singles in Albums (*Duplicates*)

- Correlation (R) = 0.7877

Linear + Residual Analysis

- Fitted squared model
- $R^2 = 0.452$
- $\beta_1 = 67.2372$
- P-value = 0.000
 - Significant
 - Can reject the null
 - Popularity single has impact on popularity album





Changes in Musical Attributes *Analysis*

Musical Attributes:

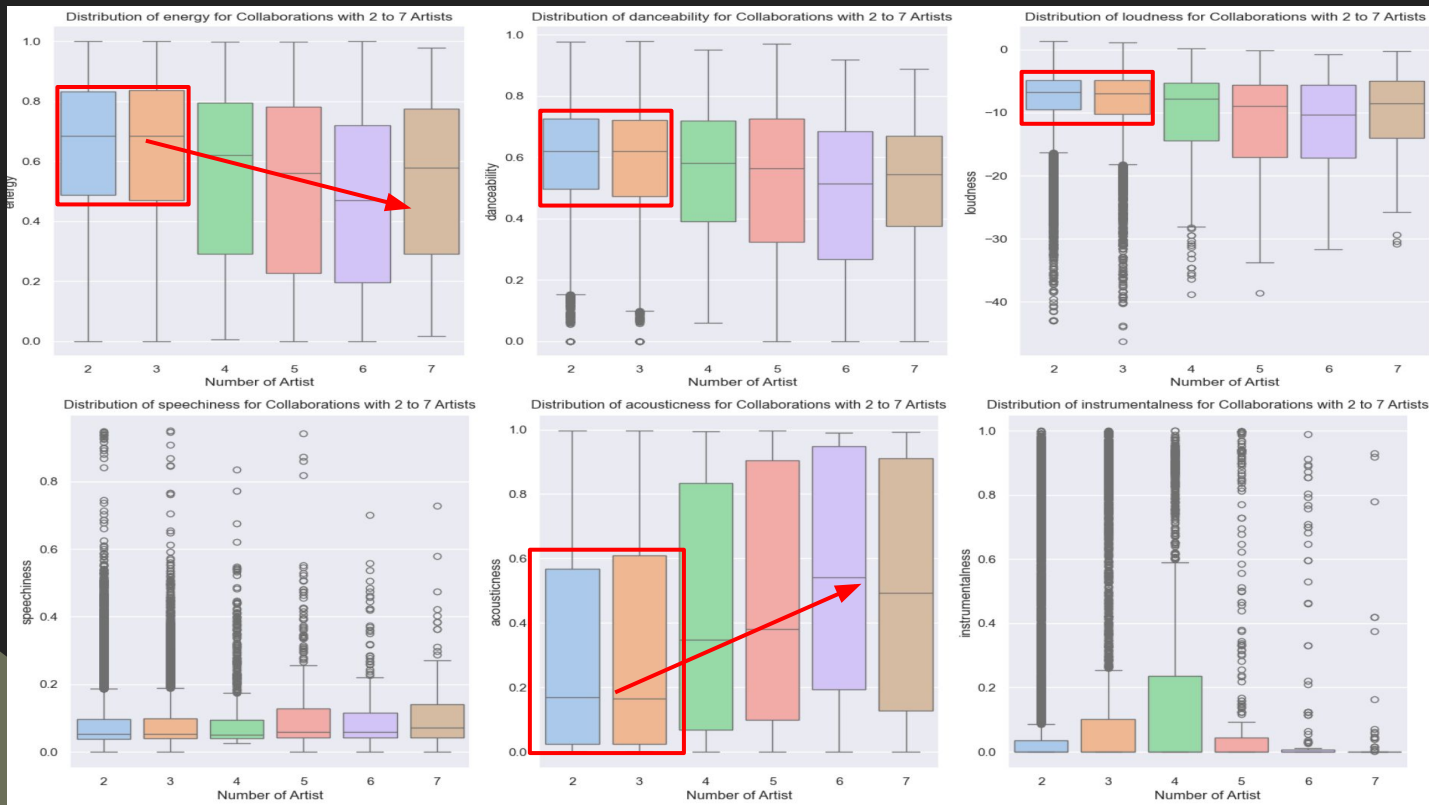
*Energy, Danceability, Loudness, Speechiness, Acousticness,
Instrumentalness, Liveness, Valence, Tempo, Duration, Time_signature*



03.2 - Analysis

Topic Two: Changes in Musical Attributes

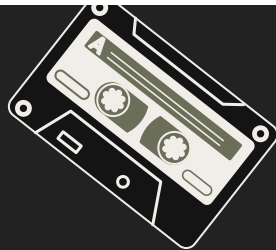
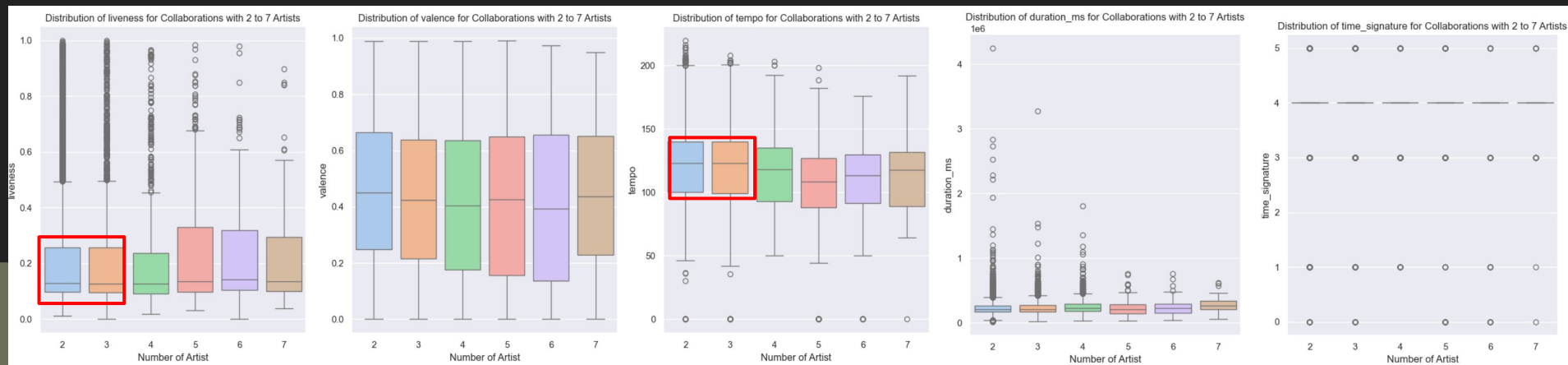
Number of Collaborators



03.2 - Analysis

Topic Two: Changes in Musical Attributes

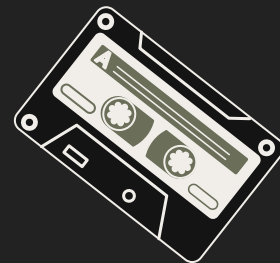
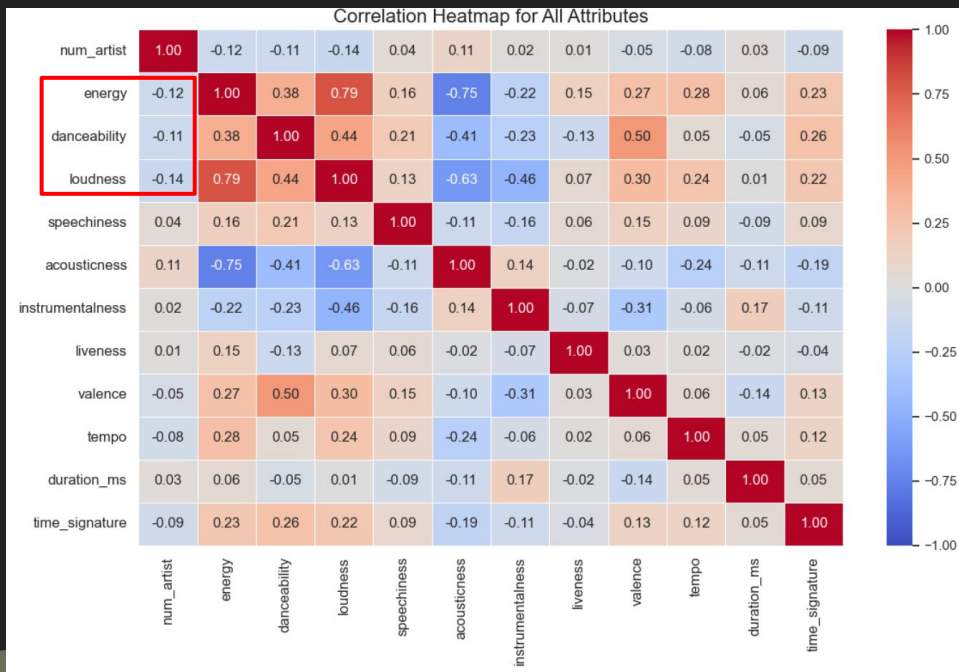
Number of Collaborators



03.2 - Analysis

Topic Two: Changes in Musical Attributes

Number of Collaborators



03.2 - Analysis

Topic Two: Changes in Musical Attributes

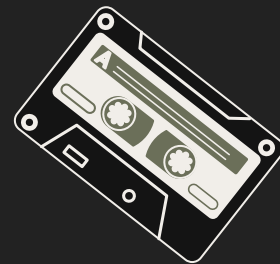
Different Artist Genre Collabs



Country singer x Pop Rapper
DIFFERENT GENRE COLLAB



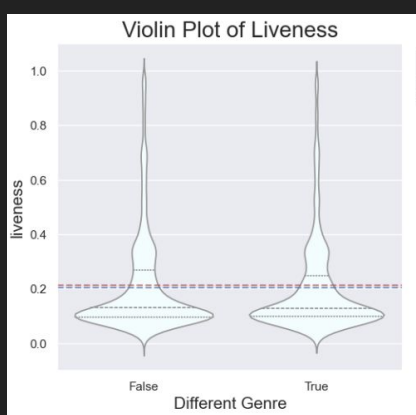
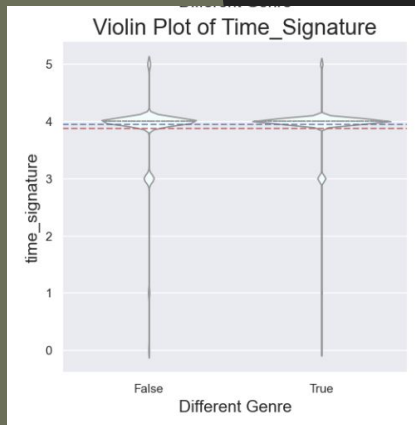
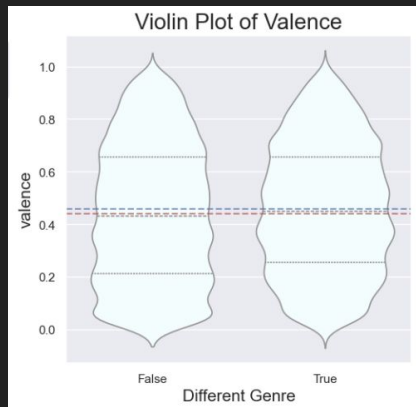
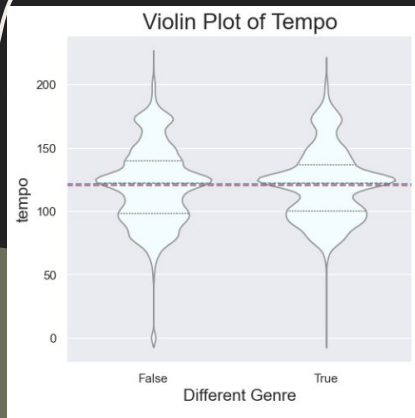
Pop singer x Pop Singer
SAME GENRE COLLAB



03.2 - Analysis

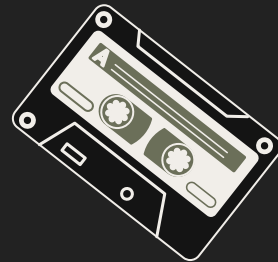
Topic Two: Changes in Musical Attributes

Different Artist Genre Collabs



- Artists =
Diff or Same Genre
- Similar variations, and
peaks
- Same average/means

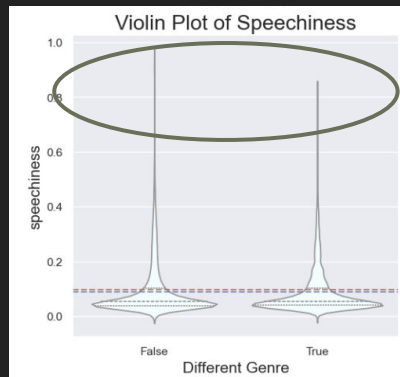
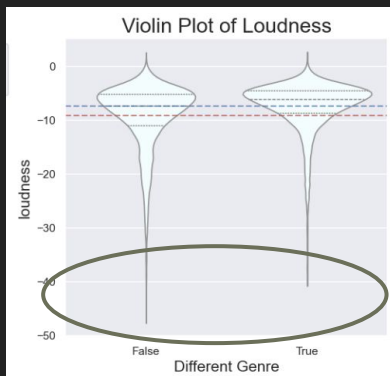
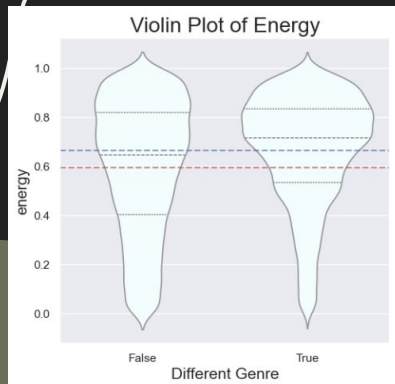
Mean by Genre
- - - Mean (False)
- - - Mean (True)



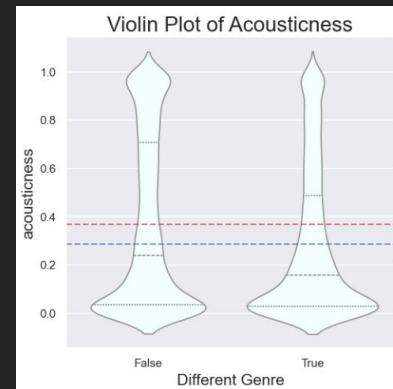
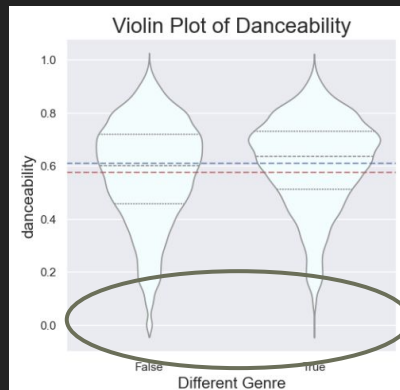
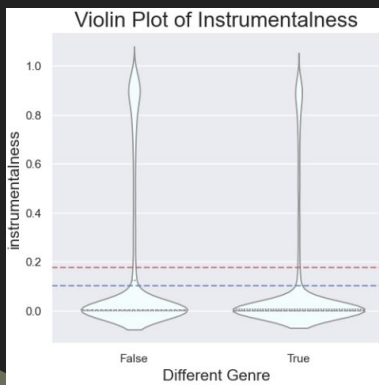
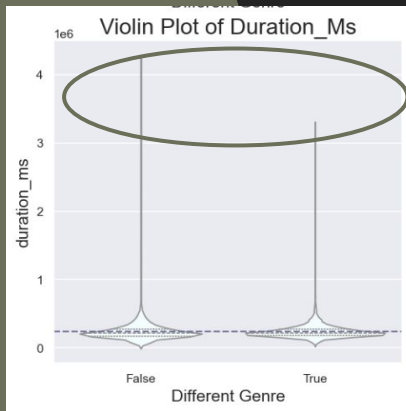
03.2 - Analysis

Topic Two: Changes in Musical Attributes

Different Artist Genre Collabs



- Different in some variations or peaks
- Different average/means



03.2 - Analysis

Topic Two: Changes in Musical Attributes

Different Artist Genre Collabs

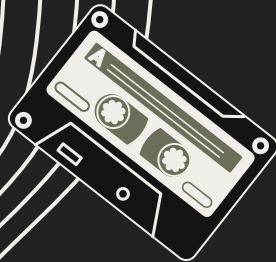
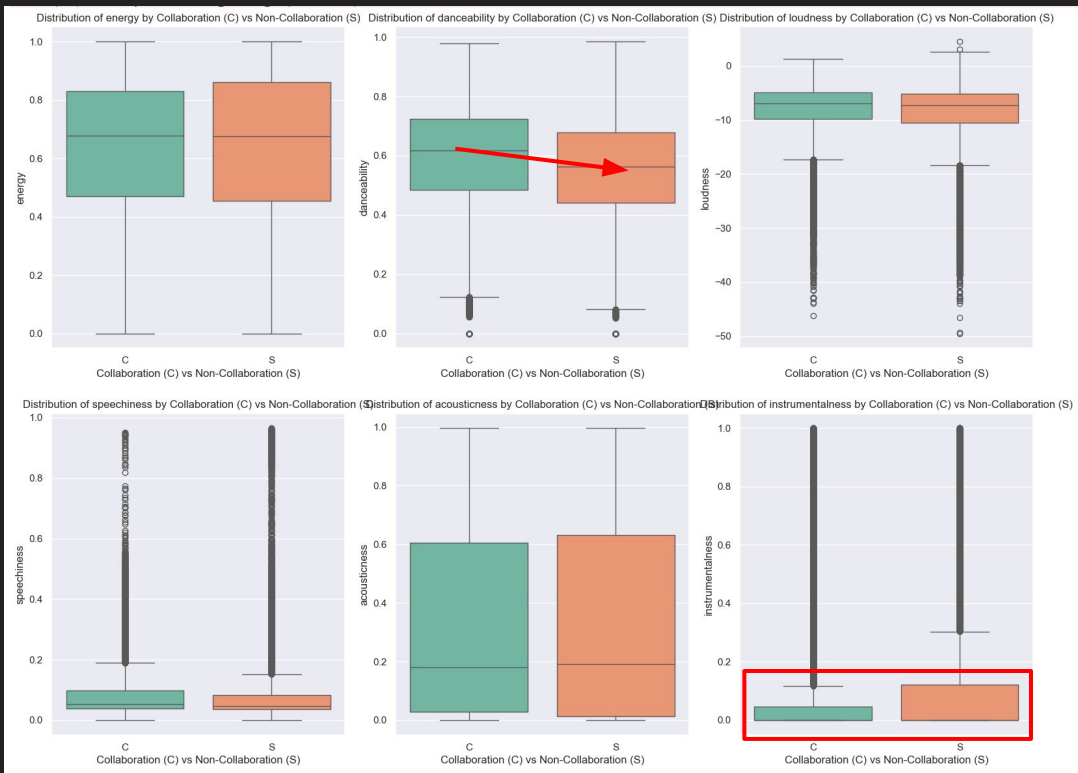
	Attribute	t-statistic	p-value	Significant
0	danceability	-20.375972	5.876504e-92	True
1	energy	-28.735192	2.806035e-180	True
2	loudness	-33.758540	2.403381e-247	True
3	speechiness	8.970029	3.052945e-19	True
4	acousticness	26.594080	7.245776e-155	True
5	instrumentalness	27.246673	2.075122e-162	True
6	liveness	5.032238	4.862782e-07	True
7	valence	-7.651911	2.011553e-14	True
8	tempo	-3.139508	1.693196e-03	True
9	time_signature	-14.794768	1.961598e-49	True
10	duration_ms	-2.377901	1.741482e-02	True

- Hypothesis:
Different genre collaborations > more extreme musical attributes
- Significant:
P-value < 5%
- Reject null hypothesis
- Mean value changes
- Inconclusive - extremes
- Need different test - compare variance

03.2 - Analysis

Topic Two: Changes in Musical Attributes

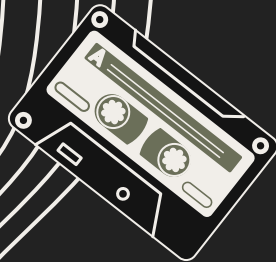
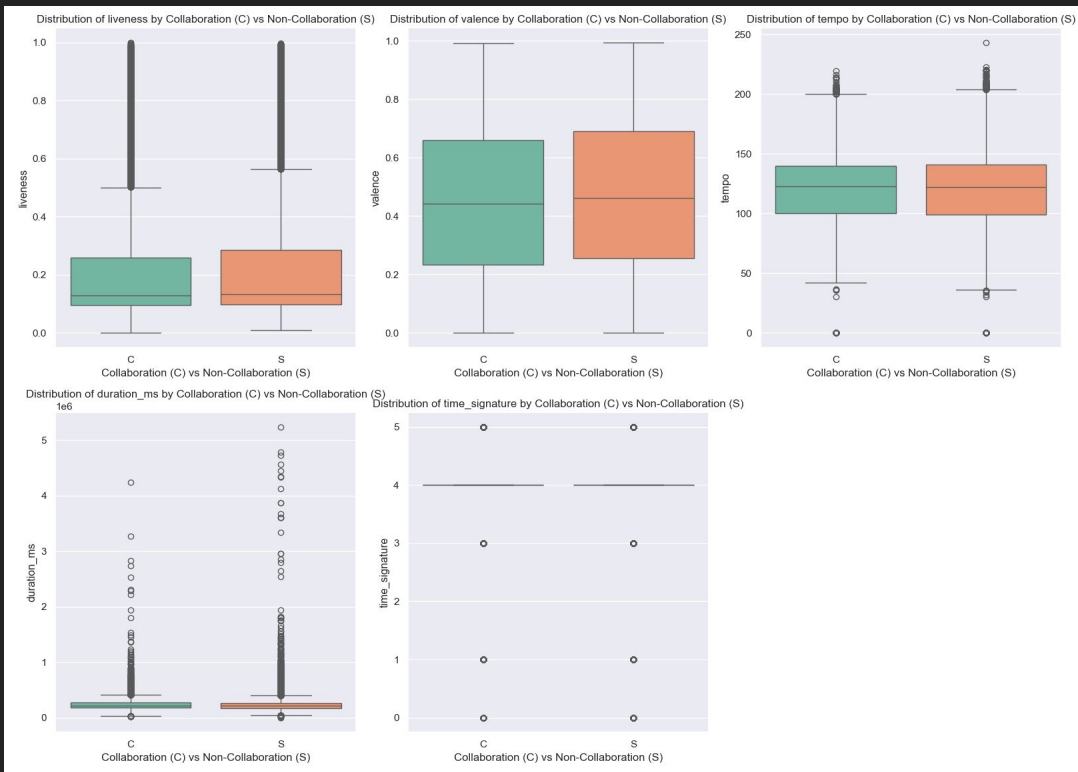
Attribute Relations to Collabs



03.2 - Analysis

Topic Two: Changes in Musical Attributes

Attribute Relations to Collabs



03.2 - Analysis

Topic Two: Changes in Musical Attributes

Attribute Relations to Collabs

- P-value = 0.000
 - Can reject the null
- Low R-squared

Regression Results for danceability:

```
=====
                        OLS Regression Results
=====
Dep. Variable:          danceability      R-squared:                0.010
Model:                  OLS              Adj. R-squared:           0.010
Method:                 Least Squares     F-statistic:              935.0
Date:                   Mon, 09 Dec 2024   Prob (F-statistic):       2.78e-204
Time:                   10:23:48          Log-Likelihood:           28681.
No. Observations:       89741            AIC:                     -5.736e+04
Df Residuals:           89739            BIC:                     -5.734e+04
Df Model:                1
Covariance Type:        nonrobust
=====
                        coef      std err      t      P>|t|      [0.025      0.975]
-----
const                0.5518      0.001    813.436    0.000      0.550      0.553
collab_binary        0.0413      0.001    30.577    0.000      0.039      0.044
=====
Omnibus:                2438.759    Durbin-Watson:           1.026
Prob(Omnibus):           0.000    Jarque-Bera (JB):        2616.621
Skew:                    -0.412    Prob(JB):                 0.00
Kurtosis:                2.857    Cond. No.                 2.48
=====
```

Notes:

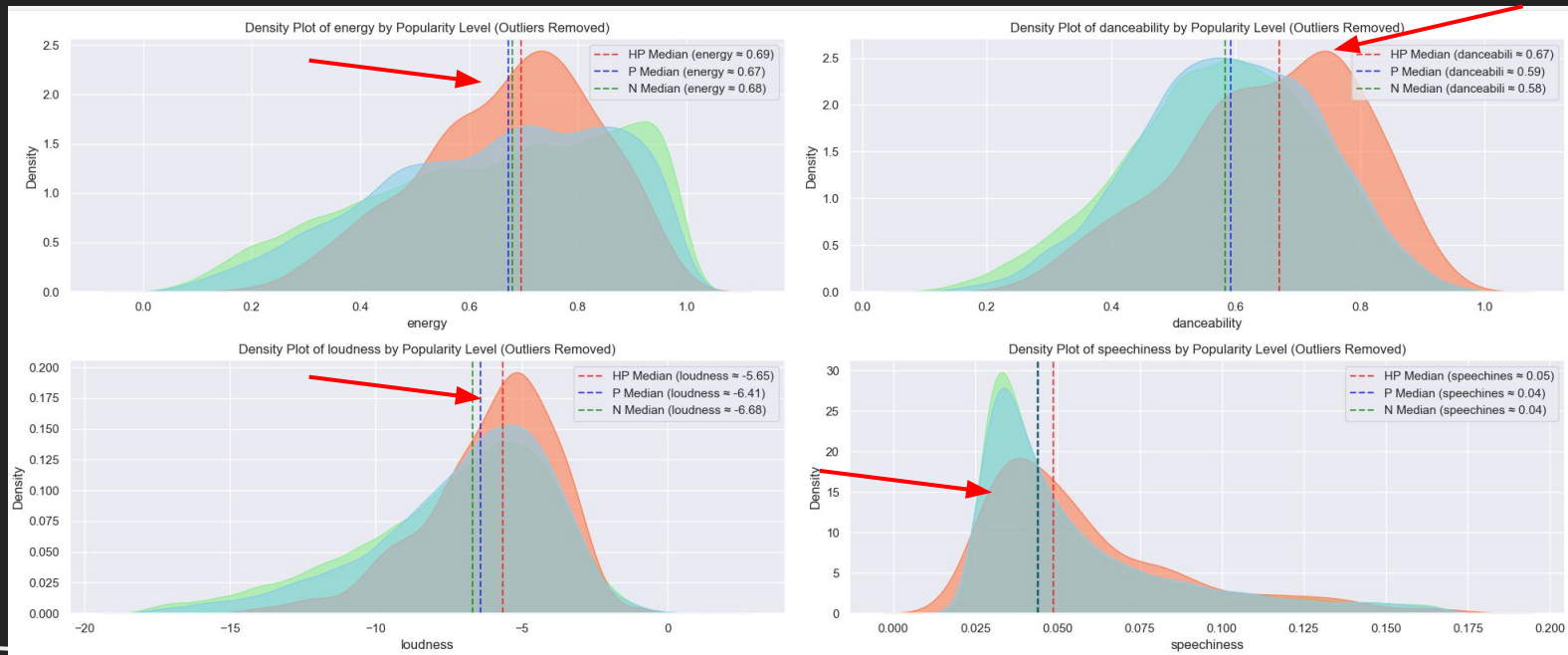
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



03.2 - Analysis

Topic Two: Changes in Musical Attributes

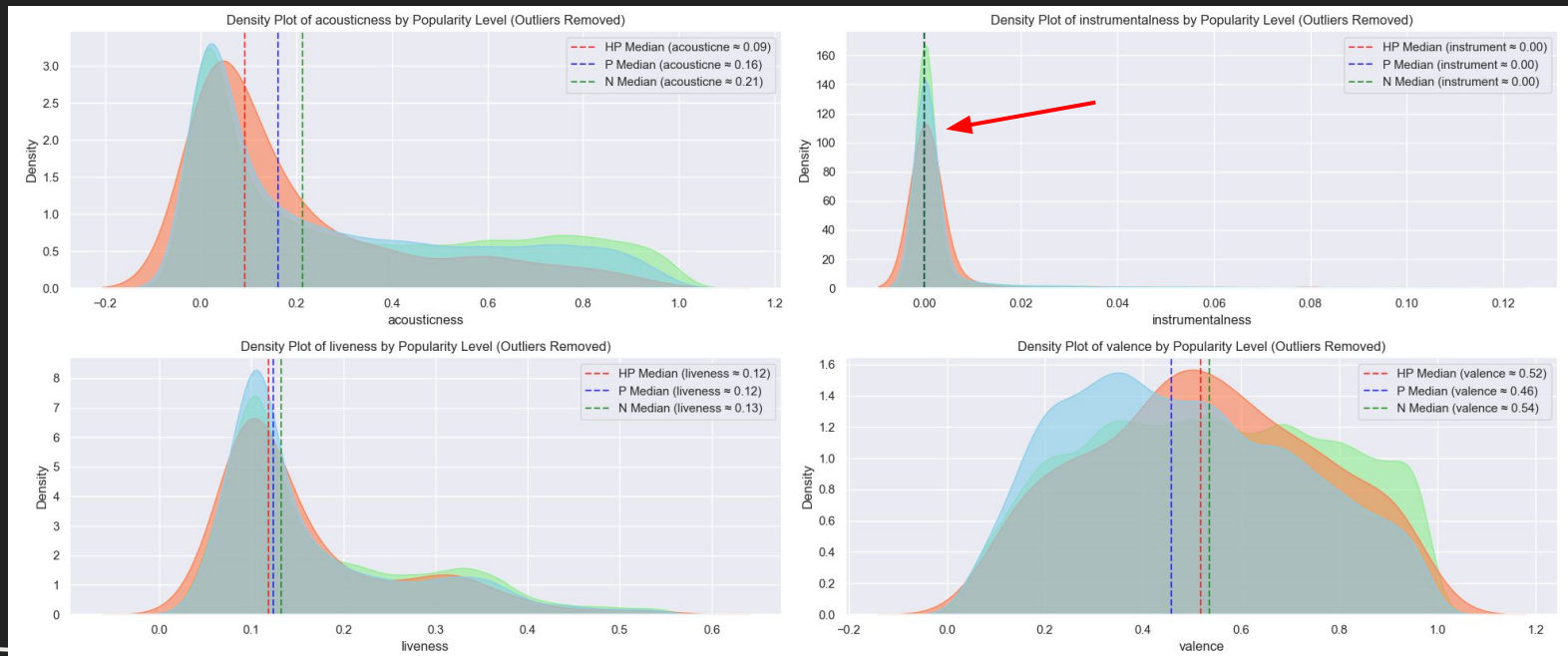
Attribute Relation to Popularity



03.2 - Analysis

Topic Two: Changes in Musical Attributes

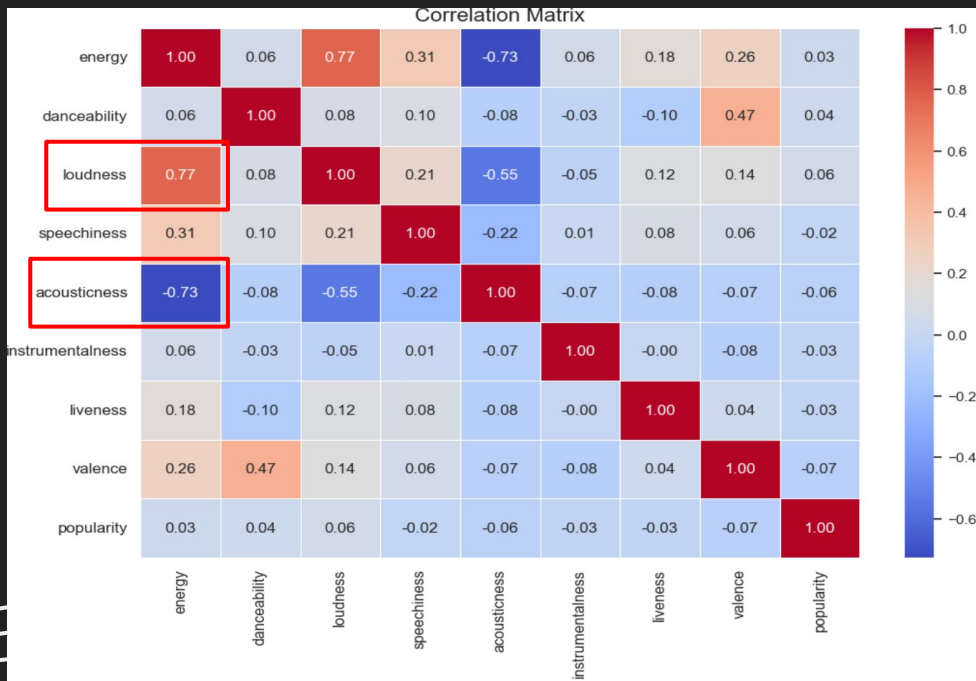
Attribute Relation to Popularity

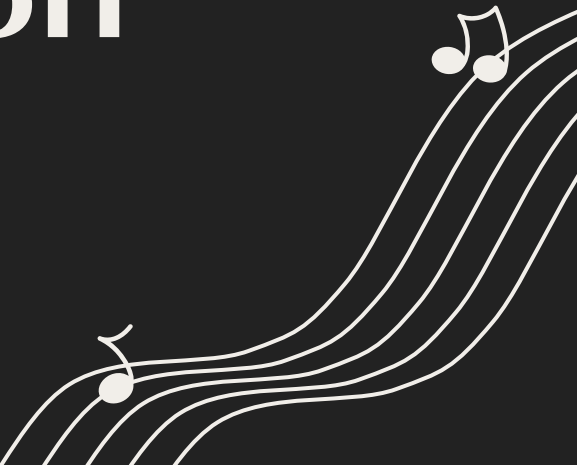


03.2 - Analysis

Topic Two: Changes in Musical Attributes

Attribute Relation to Popularity





03

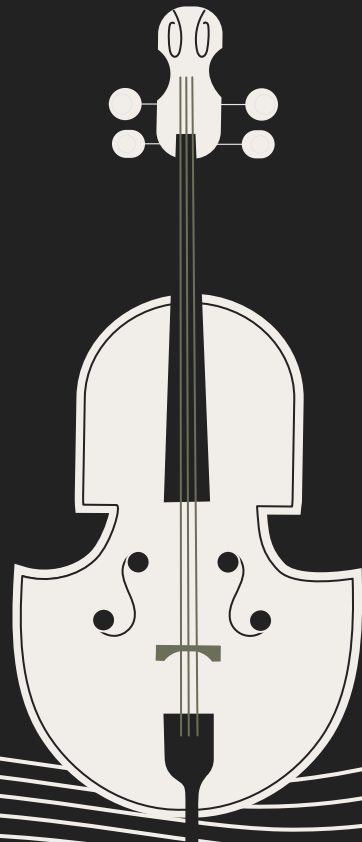
Conclusion

04.1 – Communication

Topic One: Impact of Popular Songs

How does the presence of one or more highly popular songs in an album affect the popularity and listener engagement of other tracks within the same album?

Number of highly popular songs	Yes relationship/trend found
Presence of popular vs highly popular song	Yes relationship/trend found
Consistent for genres?	Yes relationship/trend found
Impact of popular collaboration v popular solo?	Slight relationship observed

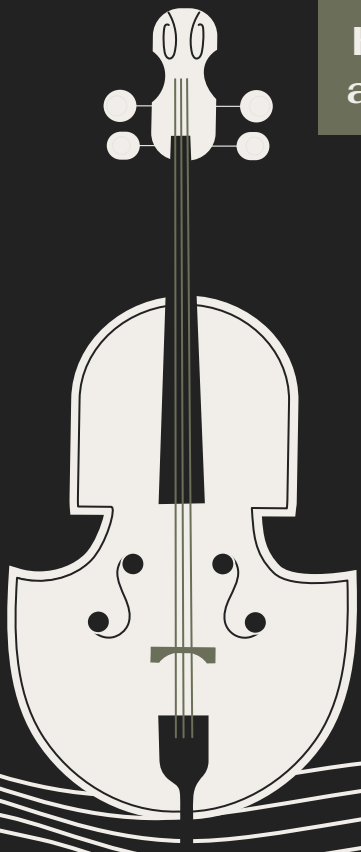


04.2 – Communication

Topic Two: Changes in Musical Attributes

How do the musical attributes of tracks change when artists collaborate compared to when they work solo?

Number of collaborators	Various changes observed
Different artist genres collaborate	Average value of musical attribute changes
Attribute R collaborations	Slight differences observed
Attribute R popularity	Some musical attributes has correlation but limited





References



Ahlgrim, C. (2024, May 3). *There are 24 artists who have been featured on a Taylor Swift song - here they all are.* Business Insider.

<https://www.businessinsider.com/taylor-swift-features-songs-with-other-artists-2023-10>

Data science steps as scientific method for big data analyze outline diagram. Pinterest. (2023, March 17).
<https://www.pinterest.com/pin/data-science-steps-as-scientific-method-for-big-data-analyze-outline-diagram--595460382003312783/>

Hatton, M. (2021, May 7). What is Jupyter?. Nightingale HQ.
<https://nightingalehq.ai/knowledgebase/glossary/what-is-jupyter/>

Fatima, M. (2023, October 22). *Understanding Spotify logo meaning, history, and evolution.* World Branding Forum. <https://brandingforum.org/general/understanding-spotify-logo-meaning-history-and-evolution/>

Mezydlo, J. (2024, July 11). *Unlikely music collaborations that worked.* Yardbarker.
https://www.yarbarker.com/entertainment/articles/unlikely_music_collaborations_that_worked/s1__38769852#slide_19

Spotify. (2024). *Community.* Spotify for Developers. <https://developer.spotify.com/community>

